



Intro to Information Security

Outcome:

1. You'll be able to understand the importance of Information Security.
2. You'll be able to understand terminologies used in Security domain.
3. . You'll be able to have deep understanding of security as opposed to just under the lock and key.



Introduction

- Traditionally we think of Security as protecting the resources under the lock and key .
- But in modern world today **Security** is a state of wellbeing (are you secure?)
- Security is all about being prepared for the unexpected (challenge) .
- And so Information Security is the collection of the following:



6.1.1 Policies

- They tell you what to do.
 - Example: Real life: ID cards
 - Example: IT world: Password policies
- *A password must have mixture of numbers, characters, special characters - minimum 8 in length



6.1.2 Procedures

They tell you how to do.

Real life example: Show the ID card to the guard before entering the premises.

IT world example:

(I) All user passwords must contain at least two (2) alphabetic and two (2) non-alphabetic character. Non-alphabetic characters include numbers (0-9) and special characters.

II. Procedure to install a package (WordPress), Linux or windows.



6.1.3 Practices

- Patterns that are proven to get the desired output.
- Real life example:
 - I. Wearing your ID at all times while in premises.
 - II. Health practices - Washing your hands , covering your face with a mask , changing tooth brush.
- IT world example:
 - Changing Passwords every 30 days, refrain from using easy password like birthdates.



6.1.4 Confidentiality

- The state of keeping or being kept secret or private.
- Confidentiality prevents the unauthorized use or disclosure of information, ensuring that only those who are authorized to access information can do so.
- In other words it is restrictions on the accessibility (who can access the data) and dissemination of the information (if you have the data and you are authorized then you must keep it to yourself).
- Real life example - generally , we do not publish our bank account balance on Facebook.
- Example in IT world - Keep your password confidential.



6.1.5 Integrity

- I. Protecting data from modification or deletion by unauthorized parties.
- II. Ensuring that when authorized people make changes , that shouldn't been made, the damage can be undone.
- Real life example:
- Students --> Sharing a Room --> Wallet on the bed --> and no money stolen - Integrity
- Example if IT world:
- Computer A --> (DATA - ABC) --> Computer B (DATA - ABC) - Integrity maintained

6.1.6 Availability



Ensuring that information is available when required.

Real life example

Your balance (You have money in your BSP account, you can withdraw it when you require cash).

IT World Example:

Sales report saved in the Sales Network folder can be accessed by the sales Managers at all times.

So in a nutshell, INFORMATION SECURITY is the collection of policies, standards, practices to ensure confidentiality, integrity and availability of information.

Quiz



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I refers to completeness and accuracy of data:

- A. Availability
- B. Confidentiality
- C. Integrity
- D. Security

Correct Answer: C

6.2 Why Information Security?



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- We need information security to reduce the risk of unauthorized information access, use, disclosure, and disruption.
- We need information security to reduce risk to a level that is acceptable to the business (management).
- We need information security to improve the way we do business.



6.3 Security – the money factor involved

1.3.1 How Much Does an Information Security Analyst make?

In developed Countries where security is paramount like USA, Australia, Japan, China,

Information Security Analysts made a median salary of \$99,730 in 2019 as per record.

The best-paid 25 percent made \$128,640 that year, while the lowest-paid 25 percent made \$75,450.

1.3.2 How much Does an Information Security Analyst make in PNG?

Average BSP offer: Base Salary K46 000 + Housing Allowance of K14 000.

Many other PNG companies would offer around the same amount.



6.4 Internet Statistics - Study from a security perspective

Check this out to see the real time internet usage in the world today:

<https://www.internetlivestats.com>

The following link below shows a real-time global view of DDoS attacks, hacking attempts, and bot assaults mitigated by Imperva security services.

<https://www.imperva.com/cyber-threat-attack-map/>